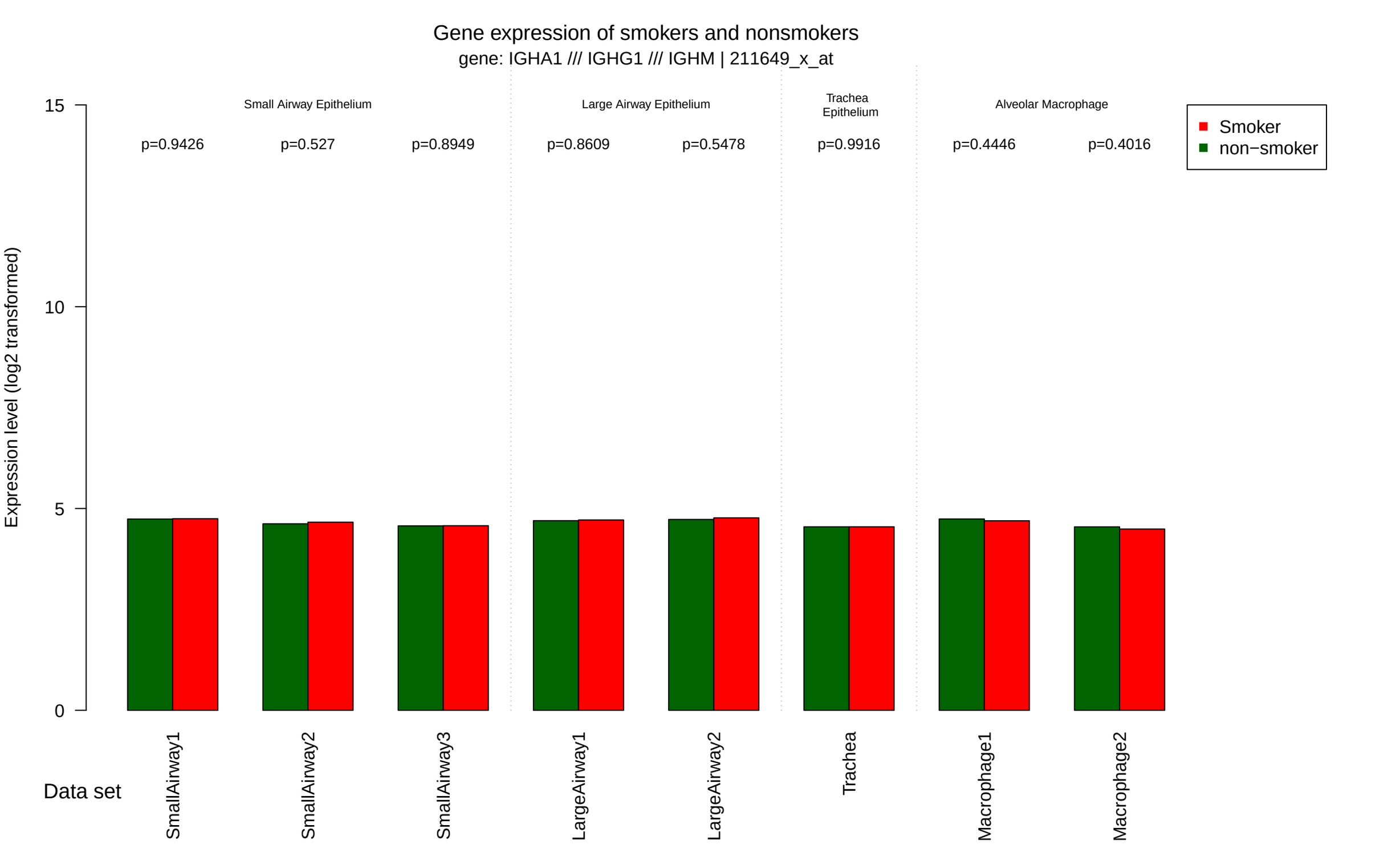
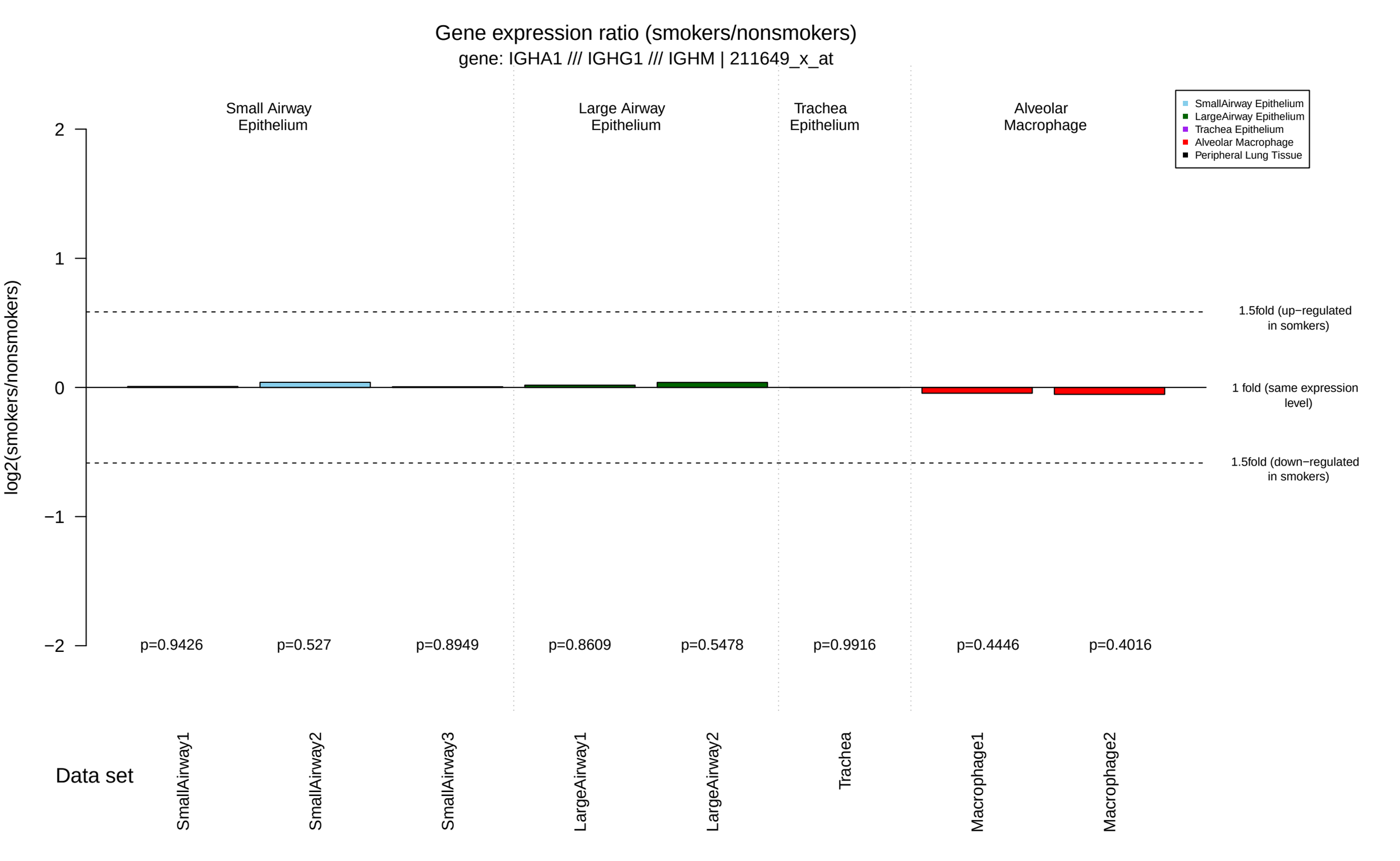
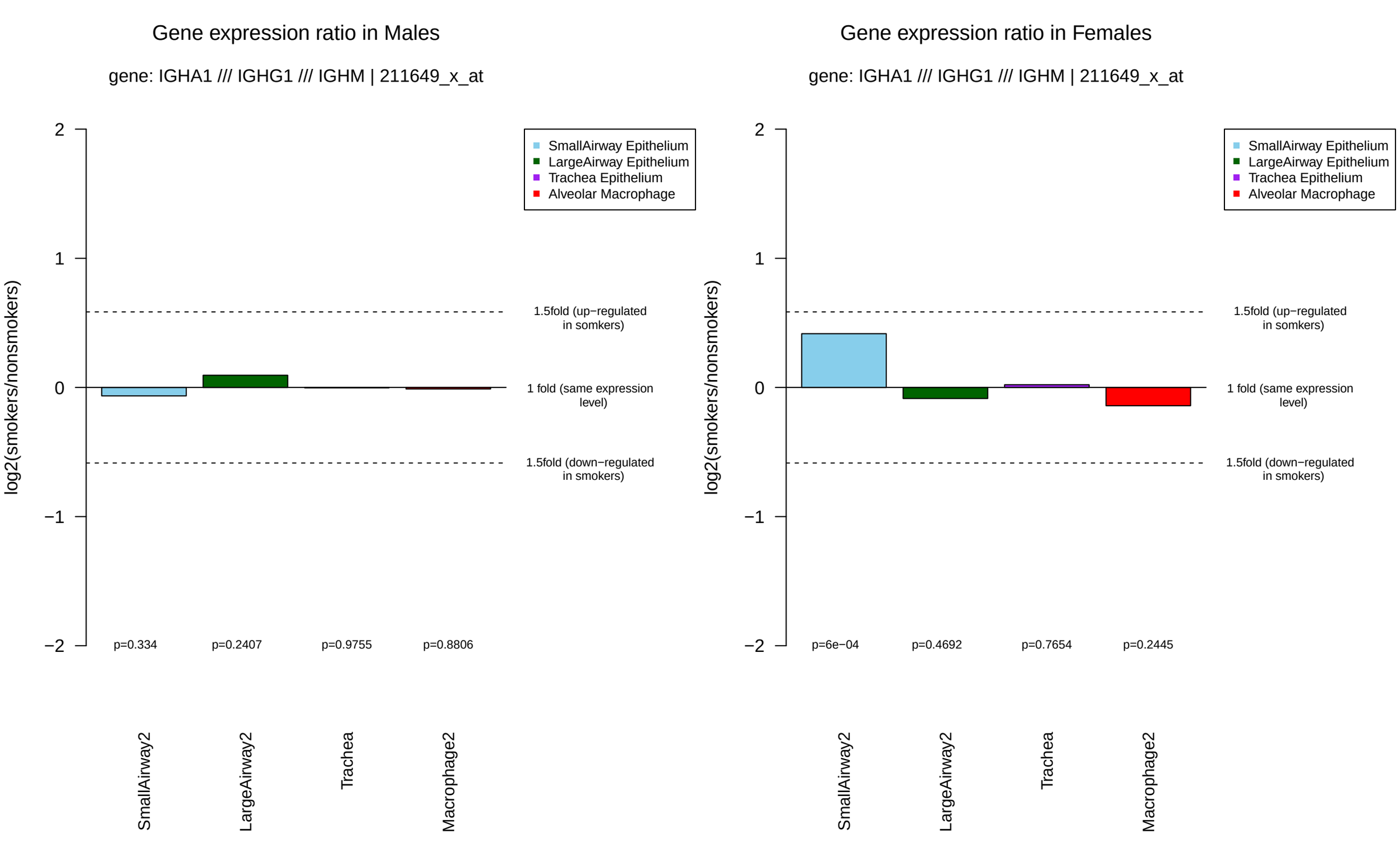
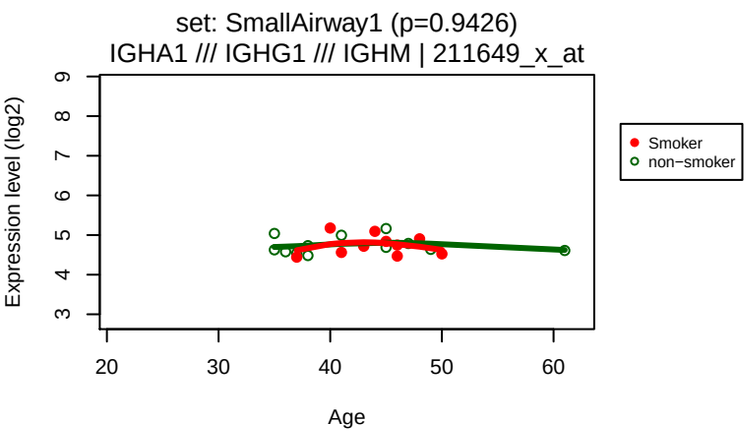
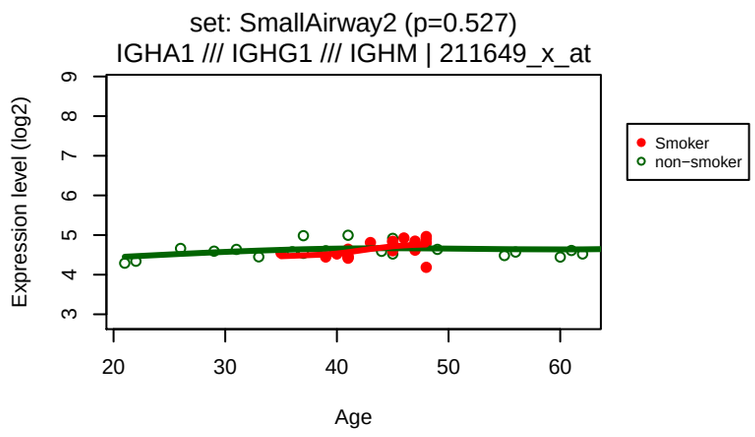
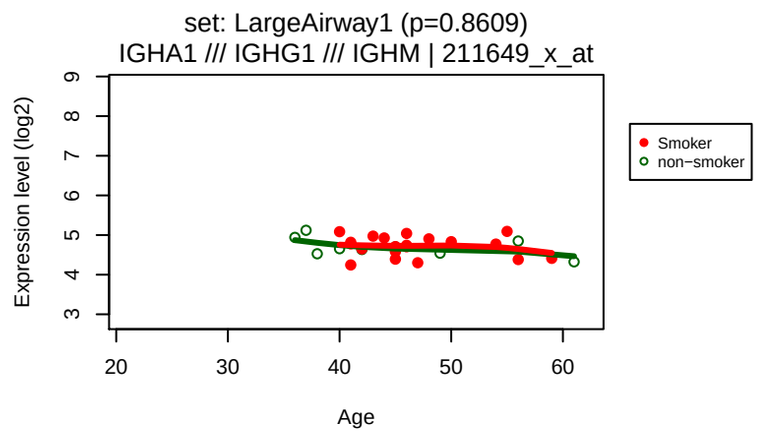

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHG1 /// IGHM | 211649_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9426
p=0.527
p=0.8949
p=0.8609
p=0.5478
p=0.9916
p=0.4446
p=0.4016
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

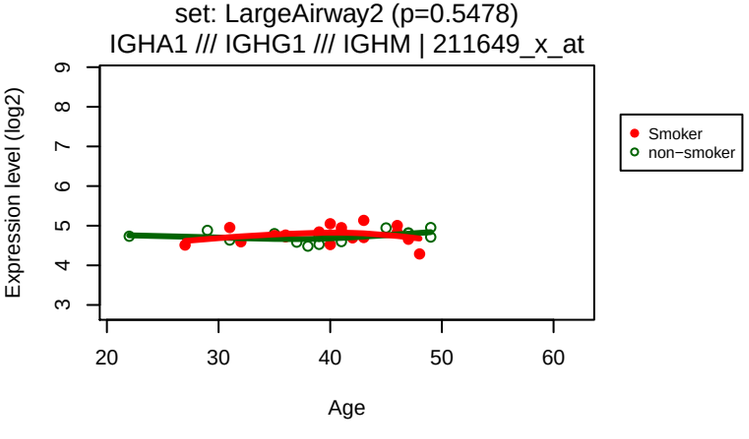

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHG1 /// IGHM | 211649_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9426
p=0.527
p=0.8949
p=0.8609
p=0.5478
p=0.9916
p=0.4446
p=0.4016
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

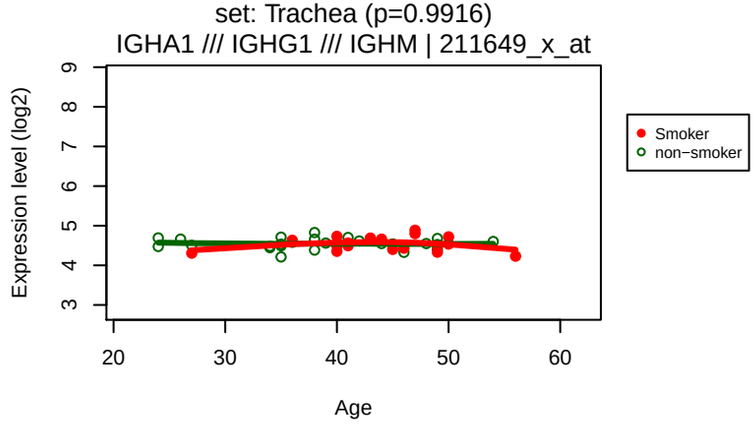

Gene expression ratio in Males
gene: IGHA1 /// IGHG1 /// IGHM | 211649_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.334
p=0.2407
p=0.9755
p=0.8806
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHA1 /// IGHG1 /// IGHM | 211649_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=6e−04
p=0.4692
p=0.7654
p=0.2445
SmallAirway2
LargeAirway2
Trachea
Macrophage2

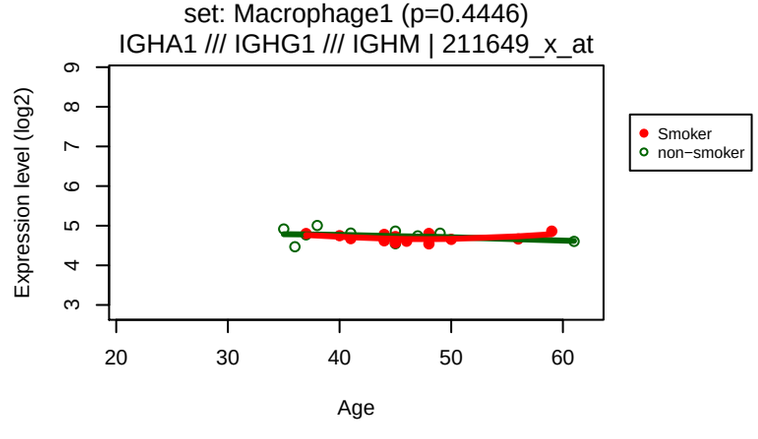

set: SmallAirway1 (p=0.9426)
IGHA1 /// IGHG1 /// IGHM | 211649_x_at
Expression level (log2)
Age
Smoker
non−smoker

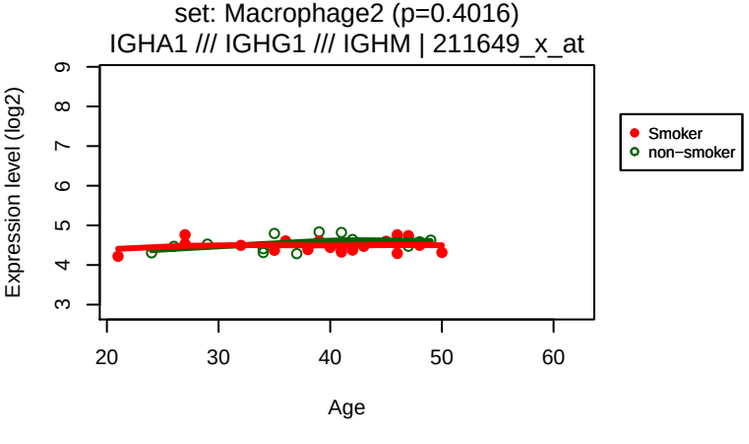

set: SmallAirway2 (p=0.527)
IGHA1 /// IGHG1 /// IGHM | 211649_x_at
Expression level (log2)
Age
Smoker
non−smoker

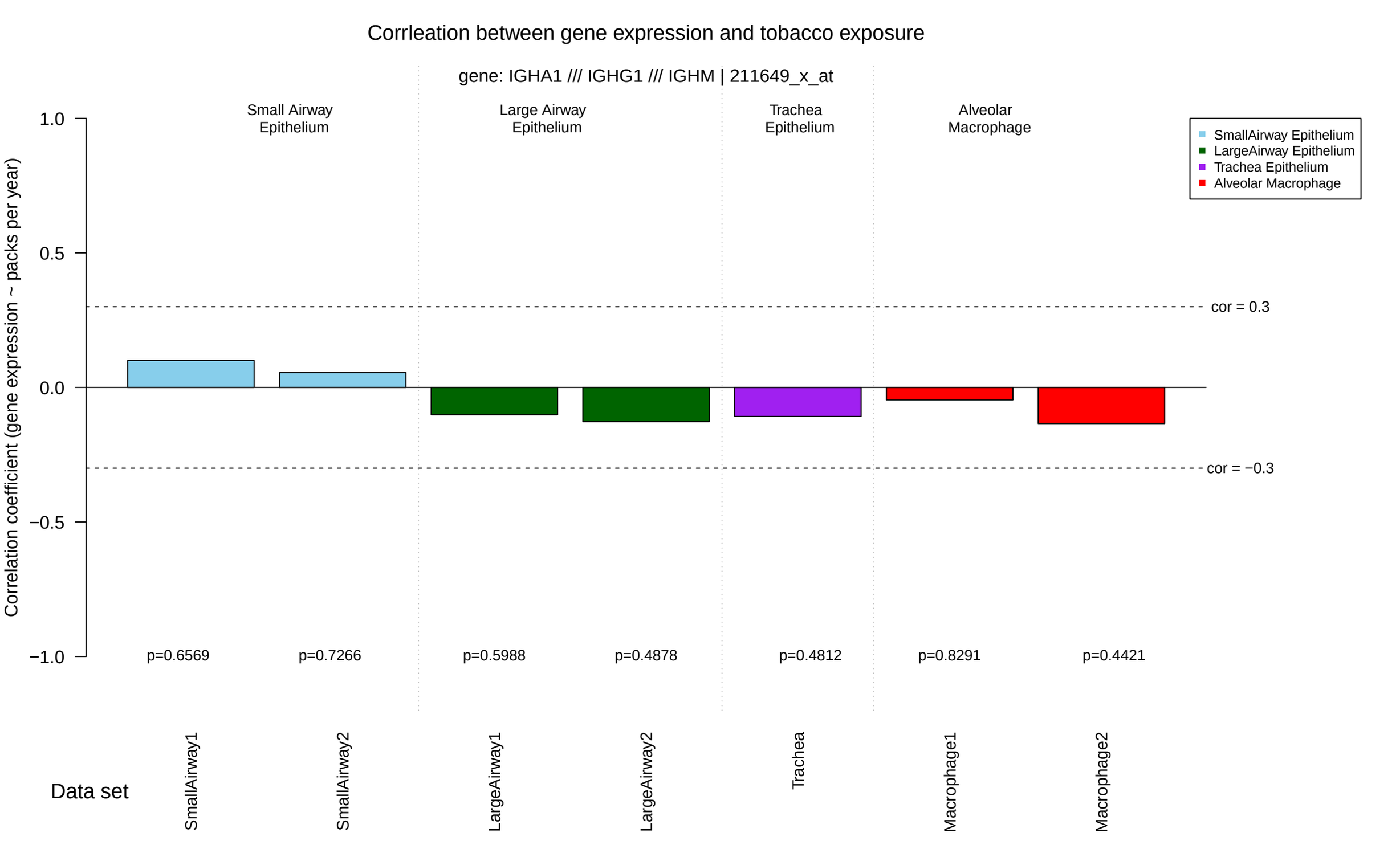

set: LargeAirway1 (p=0.8609)
IGHA1 /// IGHG1 /// IGHM | 211649_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5478)
IGHA1 /// IGHG1 /// IGHM | 211649_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9916)
IGHA1 /// IGHG1 /// IGHM | 211649_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4446)
IGHA1 /// IGHG1 /// IGHM | 211649_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4016)
IGHA1 /// IGHG1 /// IGHM | 211649_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHG1 /// IGHM | 211649_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.6569
p=0.7266
p=0.5988
p=0.4878
p=0.4812
p=0.8291
p=0.4421
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2